

A Research Principle for the AI Era

Jianmin Liao

September 28, 2026

Research in the AI era should pursue intrinsic value while building the external support needed to sustain that pursuit.

Even when intrinsic value is the sole objective, an initial phase devoted entirely to external return can be necessary for optimal research. We prove this for a linear benchmark with a sufficiently long horizon, then illustrate the optimal strategy using the same fixed linear functions in every cycle.

Here, intrinsic value means the scientific understanding or other substantive contribution that the researcher ultimately values. External return means support convertible into research funding, measured in the same units as expenditure. The distinction separates the purpose of research from the resources that sustain it.

Assumptions

Consider a fixed number n of research cycles. In cycle i , let c_i be expenditure and $p_i \in [0, 1]$ the share allocated to intrinsic work. Write I_i and E_i for intrinsic value and external return, respectively.

1. Increasing production and funding functions between linear bounds. Choose units so that $c_1 = 1$, and write

$$I_i = p_i f_i(c_i), \quad E_i = (1 - p_i) g_i(c_i), \quad c_{i+1} = h_i(E_i).$$

Each function is strictly increasing. For each family $F = f, g, h$, there are constants $0 < \ell_F < L_F$ such that

$$\ell_F x \leq F_i(x) \leq L_F x \quad (x \geq 0, 1 \leq i \leq n).$$

The bounds apply to function values; the functions may be nonlinear and may change across cycles. They bound output per unit of input without fixing curvature. The proportion enters directly as p_i or $1 - p_i$. For the final cycle, the funding update is only a convenient notation.

2. A positive and growing safety balance. Let B_i denote the cumulative funding surplus at the end of cycle i :

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

Require

$$B_i \geq \max\{\delta, \rho B_{i-1}\}, \quad \delta > 0, \quad \rho > 1.$$

The fixed floor δ establishes a safety reserve, while ρ requires it to grow. Funding is assessed at cycle boundaries; the initial expenditure is supplied as seed funding.

3. Intrinsic value as the sole objective. Choose the allocation shares to maximize

$$\sum_{i=1}^n I_i.$$

All functions are deterministic and known when the allocation is chosen. Each cycle's intrinsic value receives equal weight, and external return has value through the funding dynamics and constraints.

An exact result for the linear benchmark

For the theorem, specialize to

$$f_i(c) = c, \quad g_i(c) = bc, \quad h_i(e) = e,$$

with $0 < \delta \leq b - 1$ and $1 < \rho < b$. These inequalities make the initial reserve feasible and leave room for intrinsic work while the reserve grows. Define

$$K = \min \left\{ m \geq 1 : (b - \rho) \sum_{j=0}^{m-1} \rho^j > 1 \right\}, \quad n > K.$$

Here m is an integer. Since $b > \rho > 1$, K is finite. The horizon condition gives early external returns enough time to generate later intrinsic value.

Theorem 1. *In this linear benchmark, every optimal strategy devotes the first $n - K$ cycles entirely to external return:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

There is an optimal strategy that then meets the balance-growth constraint exactly in all remaining cycles:

$$B_i = \rho B_{i-1}, \quad p_i = 1 - \frac{1 + \rho B_{i-1}}{b(1 + B_{i-1})} \in (0, 1) \quad (n - K < i \leq n).$$

Thus, maximizing intrinsic value alone requires an initial phase of full external focus and admits a single subsequent switch to a mixture of intrinsic and external work.

A numerical example

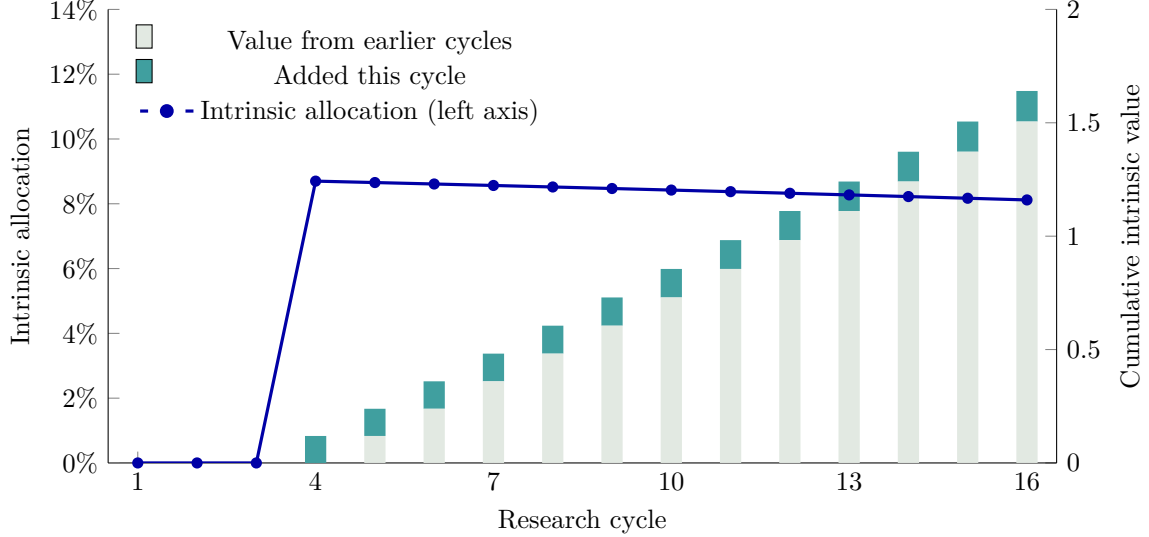


Figure 1: An optimal trajectory with the same fixed linear functions in every cycle: three cycles of full external focus, followed by thirteen cycles of mixed allocation. The line shows the intrinsic allocation share (left axis). Each full bar shows cumulative intrinsic value (right axis); its darker segment is the intrinsic value added in that cycle, and its lighter segment is the value accumulated in earlier cycles.

Use the same three linear functions in all sixteen cycles:

$$f_i(c) = c, \quad g_i(c) = 1.11c, \quad h_i(e) = e,$$

with $c_1 = 1$, $B_0 = 0$, $\delta = 0.1$, and $\rho = 1.05$. The theorem gives $K = 13$, so the optimal strategy has a short initial phase of three cycles devoted entirely to external return, followed by thirteen cycles of mixed allocation.

During cycles 1–3, $p_i = 0$ and $B_i = 1.11^i - 1$. During cycles 4–16, the balance grows by exactly 5% each cycle, and the allocation follows the theorem’s formula. The intrinsic share is 8.70% in cycle 4 and 8.12% in cycle 16. Although this share decreases slightly as the safety balance grows, the larger research budget raises the intrinsic increment from 0.1190 to 0.1348. Cumulative intrinsic value reaches 1.6403 in normalized units.

The figure displays this single optimal trajectory. No functions or coefficients change across cycles, and no averaging is used. The fixed safety floor and balance-growth requirement hold in every cycle.

Proof

For the linear benchmark, the funding identity telescopes to

$$B_i = c_{i+1} - 1, \quad c_i = 1 + B_{i-1}.$$

Consequently, at any cycle after the first, a current balance $B \geq \delta$ permits exactly the next balances B' satisfying

$$\rho B \leq B' \leq bB + b - 1.$$

The intrinsic value earned in that cycle is

$$I = 1 - \frac{1}{b} + B - \frac{B'}{b}.$$

The upper endpoint corresponds to $p = 0$; the lower endpoint meets the required balance growth exactly.

Let $V_m(B)$ be the largest total intrinsic value attainable with m cycles remaining and current balance $B \geq \delta$. We show by induction that V_m is affine in B , and denote its slope by A_m . Since $V_0 = 0$, $A_0 = 0$. The recursion is

$$V_m(B) = 1 - \frac{1}{b} + B + \max_{\rho B \leq B' \leq bB + b - 1} \left\{ V_{m-1}(B') - \frac{B'}{b} \right\}.$$

If V_{m-1} is affine, the coefficient of B' is $A_{m-1} - 1/b$. Therefore the maximizing choice is the upper endpoint when $A_{m-1} > 1/b$, and the lower endpoint when $A_{m-1} < 1/b$. At equality, either endpoint is optimal; choose the lower one. Substitution shows that V_m is affine and gives

$$A_m = \begin{cases} 1 - \rho/b + \rho A_{m-1}, & A_{m-1} \leq 1/b, \\ bA_{m-1}, & A_{m-1} > 1/b. \end{cases}$$

Until the threshold is crossed,

$$A_m = \frac{b - \rho}{b} \sum_{j=0}^{m-1} \rho^j.$$

Hence $A_{K-1} \leq 1/b < A_K$. After this crossing, the second branch keeps $A_m > 1/b$. It follows that every cycle with more than K cycles remaining has the unique optimal choice of full external focus. With at most K cycles remaining, choosing the lower endpoint is optimal.

For the first cycle, the feasible interval is $\delta \leq B_1 \leq b - 1$. Its objective has coefficient $A_{n-1} - 1/b$ on B_1 . Since $n > K$, this coefficient is strictly positive, so every optimum chooses $B_1 = b - 1$, equivalently $p_1 = 0$. The same argument applies to the subsequent cycles with more than K cycles remaining.

In the last K cycles, choose $B_i = \rho B_{i-1}$. The formula for p_i follows from $E_i = 1 + B_i = b(1 - p_i)(1 + B_{i-1})$. Since $B_{i-1} > 0$ and $1 < \rho < b$, this share lies strictly between zero and one. This proves the result. $\square$

Build external support early so that the full research trajectory can deliver more intrinsic value.